123.cpp arrayop.cpp hashingop.cpp sentinal.cpp 214.cpp 2674.cpp pointers.cpp BLACKBOX

sentinal.cpp > main()

```cpp
#include <iostream>
#include <cstring>
using namespace std;

struct Friend {
    char name[50];
    char mobile[15];
};

int main() {
    Friend phonebook[100], temp;
    int n, i, j, pos;
    char searchName[50];
    bool found = false;

    cout << "Enter number of friends: ";
    cin >> n;

    for (i = 0; i < n; i++) {
        cout << "Enter name: ";
        cin >> phonebook[i].name;
        cout << "Enter mobile number: ";
        cin >> phonebook[i].mobile;
    }

    for (i = 0; i < n - 1; i++) {
        for (j = 0; j < n - i - 1; j++) {
            if (strcmp(phonebook[j].name, phonebook[j + 1].name) > 0) {
                temp = phonebook[j];
                phonebook[j] = phonebook[j + 1];
                phonebook[j + 1] = temp;
            }
        }
    }

    cout << "Enter friend's name to search: ";
    cin >> searchName;

    if (n > 0) {
        Friend last = phonebook[n - 1];
        strcpy(phonebook[n - 1].name, searchName);

        i = 0;
        while (strcmp(phonebook[i].name, searchName) != 0) {
            i++;
        }
```

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sentinal.cpp > main()

```cpp
        i = 0;
        while (strcmp(phonebook[i].name, searchName) != 0) {
            i++;
        }

        phonebook[n - 1] = last;

        if (i < n - 1 || strcmp(phonebook[n - 1].name, searchName) == 0) {
            cout << "Name: " << phonebook[i].name << endl;
            cout << "Mobile: " << phonebook[i].mobile << endl;
            cout << "Friend found\n";
            found = true;
        }
    }

    if (!found) {
        cout << "Friend not found\n";
        cout << "Enter mobile number: ";
        cin >> temp.mobile;
        strcpy(temp.name, searchName);

        pos = 0;
        while (pos < n && strcmp(phonebook[pos].name, temp.name) < 0) {
            pos++;
        }

        for (i = n; i > pos; i--) {
            phonebook[i] = phonebook[i - 1];
        }

        phonebook[pos] = temp;
        n++;

        cout << "Friend inserted successfully\n";
    }

    cout << "\nPhone Book:\n";
    for (i = 0; i < n; i++) {
        cout << phonebook[i].name << "\t" << phonebook[i].mobile << "\n";
    }

    return 0;
}
```

Problems Output Terminal Debug Console Ports Code

```
PS C:\Users\Admin\Untitled Folder 1> cd "c:\Users\Admin\Untitled Folder 1\" ; if ($?) { g++ assignment_3.cpp -o assignment_3 } ; if ($?) { .\assignment_3 }
Enter mobile number: 4362859214
Enter friend's name to search: arnav
Name: arnav
Mobile: 1234678900
Friend found

Phone Book:
arnav   1234678900
yash    4362859214
PS C:\Users\Admin\Untitled Folder 1>
```